

Public comments: paragraph corpus and structural topic model

```
# The only block you edit. Every analyst decision is a line here.
tx <- list(
  seed = 2026L,
  para_split = "\\r?\\n+", # blank-line separated? "\\r?\\n{2,}"
  para_min_tokens = 15L, # body fragments below this merge BACKWARD
  heading_max_tokens = 12L, # headings merge FORWARD (see atomization)
  min_docfreq = 5L, # DTM vocabulary floor
  run_searchK = TRUE, # TRUE for the long K-selection run
  K_grid = seq(5, 30, 3), # widen for a larger corpus
  K = 12L, # set from the searchK diagnostics
  spline_df = 5L,
  facets_per_page = 9L, # topic trend panels per page
  workers = 16L,

  # --- LLM topic labeling (runs once, after the final STM) -----
  home = TRUE, # FALSE on the work server
  # One row per step. base_url_env and api_key_env are the NAMES of the
  # .Renviro variables holding that step's endpoint and key, so the large
  # model and the stance models can use different credentials and nothing
  # sensitive appears in this file. Ignored when home = TRUE.
  steps = tibble::tribble(
    ~step, ~base_url_env, ~api_key_env, ~model,
    "label", "COMPASS_SMALL_URL", "COMPASS_SMALL_KEY", "MODEL-FOR-LABELING",
    "manager", "COMPASS_LARGE_URL", "COMPASS_LARGE_KEY", "MODEL-FOR-MANAGER")
)

suppressPackageStartupMessages({
  library(here); library(stm); library(quanteda); library(tidytext)
  library(splines)
  library(furrr); library(viridis)
  library(tidyverse) # tidyverse LAST so its verbs win
})
theme_set(theme_minimal(base_size = 12))
# Tables must not silently lose rows in the rendered PDF: knitr truncates long
# tables by default, and a 25-topic summary is past that limit.
options(knitr.table.max.print = Inf, knitr.kable.max_rows = Inf,
  max.print = 1e6)
set.seed(tx$seed)
# All paths are anchored with here(), so chunks run the same whether you
# render from the project root or run them interactively from code/.
# Layout: here() is the text_mining folder, holding code/, input/, outputs/.
```

```

`%||` <- function(x, y) if (is.null(x)) y else x
p_out <- function(f) here::here("outputs", f)
source(here::here("code", "llm_source.R"))
if (isTRUE(tx$home)) {
  source(here::here("code", "llm_home_openrouter.R"))
  tx$steps$model <- home_models[tx$steps$step]
}
dir.create(here::here("outputs"), showWarnings = FALSE, recursive = TRUE)

# Checkpoints: compute once, reload after. Delete the file to re-fit.
cache_rds <- function(path, expr) {
  if (file.exists(path)) return(readRDS(path))
  x <- force(expr); saveRDS(x, path); x
}
cache_csv <- function(path, expr) {
  if (file.exists(path)) return(readr::read_csv(path, show_col_types = FALSE))
  x <- force(expr); readr::write_csv(x, path); x
}

```

Corpus

The default corpus is `quanteda::data_corpus_inaugural`: 60 US inaugural addresses, 1789 to 2025, with `Year` already carried as a docvar. It ships with `quanteda`, so there is no extra dependency, and it is a real corpus of multi-paragraph documents rather than a simulation, which means the atomizer, the topic model, and the LLM codebook all run against text that behaves like text.

Two properties worth knowing before you read anything into the results. It yields about 1,874 paragraphs, roughly a tenth of a production corpus, so `tx$K` belongs in the low teens rather than the mid twenties. And it has **one document per year**, so document clusters and year strata coincide; that is fine here, but `Script 2`'s cluster-robust variance cannot be exercised on it, since your production corpus has many comments per year and this one has exactly one.

AT WORK replace this chunk with the read of your own table. Everything downstream is unchanged; the validation gate below runs on whatever filled `corpus_raw`.

```

corpus_raw <- tibble(
  atom_id = quanteda::docnames(quanteda::data_corpus_inaugural),
  Year = quanteda::docvars(quanteda::data_corpus_inaugural, "Year"),
  body_english = as.character(quanteda::data_corpus_inaugural))

# AT WORK, this chunk becomes one line:
# corpus_raw <- readr::read_csv(here("input", "YOUR_FILE.csv"))
# Required columns: atom_id (document id), Year (integer), body_english (text).

```

Corpus validation (keep this at work)

Stops the render with a named error on a bad read, which is cheaper than debugging a silent failure five chunks later.

```
miss <- setdiff(c("atom_id", "Year", "body_english"), names(corpus_raw))
if (length(miss)) stop("corpus_raw missing column(s): ",
                      str_c(miss, collapse = ", "))
stopifnot(
  "atom_id must be unique"      = !anyDuplicated(corpus_raw$atom_id),
  "Year must be integer-valued" = all(corpus_raw$Year ==
                                     as.integer(corpus_raw$Year)),
  "body_english must be character" = is.character(corpus_raw$body_english))
n0 <- nrow(corpus_raw)
corpus_raw <- corpus_raw |>
  dplyr::filter(!is.na(Year), !is.na(body_english),
               str_squish(body_english) != "")
glue::glue("Validation: kept {nrow(corpus_raw)} of {n0} comments, ",
           "{min(corpus_raw$Year)}-{max(corpus_raw$Year)}")
```

Validation: kept 60 of 60 comments, 1789-2025

Newline diagnostic

The atomizer can only split on what the source uses to mark paragraphs, and Excel-origin text (in-cell breaks survive as \n) carries three conventions needing three responses. Rules: mostly line breaks present, keep the default; blank lines dominant **and** the visible sample shows single breaks used as soft wraps, set `tx$para_split = "\\r?\n{2,}"`; almost no newlines, stop, since sentence chunking changes the measurement unit and is a design decision.

```
corpus_raw |>
  dplyr::mutate(b = str_replace_all(body_english, "\\r\\n", "\\n")) |>
  dplyr::summarise(comments = dplyr::n(),
                  zero_newlines = sum(str_count(b, "\\n") == 0),
                  has_blank_lines = sum(str_count(b, "\\n{2,}") > 0),
                  median_breaks = median(str_count(b, "\\n"))) |>
  tbl_paged(caption = "Line-break conventions; apply the decision rules above
before the long render.")
```

Line-break conventions; apply the decision rules above before the long render.

comments	zero_newlines	has_blank_lines	median_breaks
60	0	60	51

```
corpus_raw |> dplyr::slice_sample(n = 2) |> dplyr::pull(body_english) |>
  str_replace_all("\r?\n", "<NL>\n") |>
  str_c(collapse = "\n\n==== NEXT ==== \n\n") |> writeLines()
```

My fellow-citizens, when we assembled here on the 4th of March, 1897, there was great anxiety with regard to our currency and credit. None exists now. Then our Treasury receipts were inadequate to meet the current obligations of the Government. Now they are sufficient for all public needs, and we have a surplus instead of a deficit. Then I felt constrained to convene the Congress in extraordinary session to devise revenues to pay the ordinary expenses of the Government. Now I have the satisfaction to announce that the Congress just closed has reduced taxation in the sum of \$41,000,000. Then there was deep solicitude because of the long depression in our manufacturing, mining, agricultural, and mercantile industries and the consequent distress of our laboring population. Now every avenue of production is crowded with activity, labor is well employed, and American products find good markets at home and abroad. Our diversified productions, however, are increasing in such unprecedented volume as to admonish us of the necessity of still further enlarging our foreign markets by broader commercial relations. For this purpose reciprocal trade arrangements with other nations should in liberal spirit be carefully cultivated and promoted. The national verdict of 1896 has for the most part been executed. Whatever remains unfulfilled is a continuing obligation resting with undiminished force upon the Executive and the Congress. But fortunate as our condition is, its permanence can only be assured by sound business methods and strict economy in national administration and legislation. We should not permit our great prosperity to lead us to reckless ventures in business or profligacy in public expenditures. While the Congress determines the objects and the sum of appropriations, the officials of the executive departments are responsible for honest and faithful disbursement, and it should be their constant care to avoid waste and extravagance. Honesty, capacity, and industry are nowhere more indispensable than in public employment. These should be fundamental requisites to original appointment and the surest guaranties against removal. Four years ago we stood on the brink of war without the people knowing it and without any preparation or effort at preparation for the impending peril. I did all that in honor could be done to avert the war, but without avail. It became inevitable; and the Congress at its first regular session, without party division, provided money in anticipation of the crisis and in preparation to meet it. It came. The result was signally favorable to American arms and in the highest degree honorable to the Government. It imposed upon us obligations from which we cannot escape and from which it would be dishonorable to seek escape. We are now at peace with the world, and it is my fervent prayer that if differences arise between us and other powers they may be settled by peaceful arbitration and that hereafter we may be spared the horrors of war. Intrusted by the people for a second time with the office of President, I enter upon its administration appreciating the great responsibilities which attach to this renewed honor and commission, promising unreserved devotion on my part to their faithful discharge and reverently invoking for my guidance the direction and favor of Almighty God. I should shrink from the duties this day assumed if I did not feel that in their performance I should have the co-operation of the wise and patriotic men of all parties. It encourages me for the great task which I now undertake to believe that those who voluntarily committed to me the trust im-

posed upon the Chief Executive of the Republic will give to me generous support in my duties to “preserve, protect, and defend, the Constitution of the United States” and to “care that the laws be faithfully executed.” The national purpose is indicated through a national election. It is the constitutional method of ascertaining the public will. When once it is registered it is a law to us all, and faithful observance should follow its decrees. Strong hearts and helpful hands are needed, and, fortunately, we have them in every part of our beloved country. We are reunited. Sectionalism has disappeared. Division on public questions can no longer be traced by the war maps of 1861. These old differences less and less disturb the judgment. Existing problems demand the thought and quicken the conscience of the country, and the responsibility for their presence, as well as for their righteous settlement no more upon me than upon you. There are some national questions in the solution of which patriotism should exclude partisanship. Magnifying their difficulties will not take them off our hands nor facilitate their adjustment. Distrust of the capacity, integrity, and high purposes of the American people will not be an inspiring theme for future political contests. Dark pictures and gloomy forebodings are worse than useless. These only becloud, they do not help to point the way of safety and honor. “Hope maketh not ashamed.” The prophets of evil were not the builders of the Republic, nor in its crises since have they saved or served it. The faith of the fathers was a mighty force in its creation, and the faith of their descendants has wrought its progress and furnished its defenders. They are obstructionists who despair, and who would destroy confidence in the ability of our people to solve wisely and for civilization the mighty problems resting upon them. The American people, intrenched in freedom at home, take their love for it with them wherever they go, and they reject as mistaken and unworthy the doctrine that we lose our own liberties by securing the enduring foundations of liberty to others. Our institutions will not deteriorate by extension, and our sense of justice will not abate under tropic suns in distant seas. As heretofore, so hereafter will the nation demonstrate its fitness to administer any new estate which events devolve upon it, and in the fear of God will “take occasion by the hand and make the bounds of freedom wider yet.” If there are those among us who would make our way more difficult, we must not be disheartened, but the more earnestly dedicate ourselves to the task upon which we have rightly entered. The path of progress is seldom smooth. New things are often found hard to do. Our fathers found them so. We find them so. They are inconvenient. They cost us something. But are we not made better for the effort and sacrifice, and are not those we serve lifted up and blessed? We will be consoled, to, with the fact that opposition has confronted every onward movement of the Republic from its opening hour until now, but without success. The Republic has marched on and on, and its step has exalted freedom and humanity. We are undergoing the same ordeal as did our predecessors nearly a century ago. We are following the course they blazed. They triumphed. Will their successors falter and plead organic impotency in the nation? Surely after 125 years of achievement for mankind we will not now surrender our equality with other powers on matters fundamental and essential to nationality. With no such purpose was the nation created. In no such spirit has it developed its full and independent sovereignty. We adhere to the principle of equality among ourselves, and by no act of ours will we assign to ourselves a subordinate rank in the family of nations. My fellow-citizens, the public events of the past four years have gone into history. They are too near to justify recital. Some of them were unforeseen; many of them momentous and far-reaching in their consequences to ourselves and our relations with the rest of the world. The part which the United States bore so honorably in the thrilling scenes in China,

while new to American life, has been in harmony with its true spirit and best traditions, and in dealing with the results its policy will be that of moderation and fairness. We face at this moment a most important question that of the future relations of the United States and Cuba. With our near neighbors we must remain close friends. The declaration of the purposes of this Government in the resolution of April 20, 1898, must be made good. Ever since the evacuation of the island by the army of Spain, the Executive, with all practicable speed, has been assisting its people in the successive steps necessary to the establishment of a free and independent government prepared to assume and perform the obligations of international law which now rest upon the United States under the treaty of Paris. The convention elected by the people to frame a constitution is approaching the completion of its labors. The transfer of American control to the new government is of such great importance, involving an obligation resulting from our intervention and the treaty of peace, that I am glad to be advised by the recent act of Congress of the policy which the legislative branch of the Government deems essential to the best interests of Cuba and the United States. The principles which led to our intervention require that the fundamental law upon which the new government rests should be adapted to secure a government capable of performing the duties and discharging the functions of a separate nation, of observing its international obligations of protecting life and property, insuring order, safety, and liberty, and conforming to the established and historical policy of the United States in its relation to Cuba. The peace which we are pledged to leave to the Cuban people must carry with it the guaranties of permanence. We became sponsors for the pacification of the island, and we remain accountable to the Cubans, no less than to our own country and people, for the reconstruction of Cuba as a free commonwealth on abiding foundations of right, justice, liberty, and assured order. Our enfranchisement of the people will not be completed until free Cuba shall "be a reality, not a name; a perfect entity, not a hasty experiment bearing within itself the elements of failure." While the treaty of peace with Spain was ratified on the 6th of February, 1899, and ratifications were exchanged nearly two years ago, the Congress has indicated no form of government for the Philippine Islands. It has, however, provided an army to enable the Executive to suppress insurrection, restore peace, give security to the inhabitants, and establish the authority of the United States throughout the archipelago. It has authorized the organization of native troops as auxiliary to the regular force. It has been advised from time to time of the acts of the military and naval officers in the islands, of my action in appointing civil commissions, of the instructions with which they were charged, of their duties and powers, of their recommendations, and of their several acts under executive commission, together with the very complete general information they have submitted. These reports fully set forth the conditions, past and present, in the islands, and the instructions clearly show the principles which will guide the Executive until the Congress shall, as it is required to do by the treaty, determine "the civil rights and political status of the native inhabitants." The Congress having added the sanction of its authority to the powers already possessed and exercised by the Executive under the Constitution, thereby leaving with the Executive the responsibility for the government of the Philippines, I shall continue the efforts already begun until order shall be restored throughout the islands, and as fast as conditions permit will establish local governments, in the formation of which the full co-operation of the people has been already invited, and when established will encourage the people to administer them. The settled purpose, long ago proclaimed, to afford the inhabitants of the islands self-government as fast as they were ready for it will be pursued with

earnestness and fidelity. Already something has been accomplished in this direction. The Government's representatives, civil and military, are doing faithful and noble work in their mission of emancipation and merit the approval and support of their countrymen. The most liberal terms of amnesty have already been communicated to the insurgents, and the way is still open for those who have raised their arms against the Government for honorable submission to its authority. Our countrymen should not be deceived. We are not waging war against the inhabitants of the Philippine Islands. A portion of them are making war against the United States. By far the greater part of the inhabitants recognize American sovereignty and welcome it as a guaranty of order and of security for life, property, liberty, freedom of conscience, and the pursuit of happiness. To them full protection will be given. They shall not be abandoned. We will not leave the destiny of the loyal millions the islands to the disloyal thousands who are in rebellion against the United States. Order under civil institutions will come as soon as those who now break the peace shall keep it. Force will not be needed or used when those who make war against us shall make it no more. May it end without further bloodshed, and there be ushered in the reign of peace to be made permanent by a government of liberty under law!

==== NEXT ====

Vice President Biden, Mr. Chief Justice, Members of the United States Congress, distinguished guests, and fellow citizens: Each time we gather to inaugurate a President we bear witness to the enduring strength of our Constitution. We affirm the promise of our democracy. We recall that what binds this Nation together is not the colors of our skin or the tenets of our faith or the origins of our names. What makes us exceptional - what makes us American - is our allegiance to an idea articulated in a declaration made more than two centuries ago: We hold these truths to be self-evident, that all men are created equal; that they are endowed by their Creator with certain unalienable rights; that among these are life, liberty, and the pursuit of happiness. Today we continue a never-ending journey to bridge the meaning of those words with the realities of our time. For history tells us that while these truths may be self-evident, they've never been self-executing; that while freedom is a gift from God, it must be secured by His people here on Earth. The patriots of 1776 did not fight to replace the tyranny of a king with the privileges of a few or the rule of a mob. They gave to us a republic, a government of and by and for the people, entrusting each generation to keep safe our founding creed. And for more than 200 years, we have. Through blood drawn by lash and blood drawn by sword, we learned that no union founded on the principles of liberty and equality could survive half-slave and half-free. We made ourselves anew, and vowed to move forward together. Together, we determined that a modern economy requires railroads and highways to speed travel and commerce, schools and colleges to train our workers. Together, we discovered that a free market only thrives when there are rules to ensure competition and fair play. Together, we resolved that a great nation must care for the vulnerable and protect its people from life's worst hazards and misfortune. Through it all, we have never relinquished our skepticism of central authority nor have we succumbed to the fiction that all society's ills can be cured through government alone. Our celebration of initiative and enterprise, our insistence on hard work and personal responsibility, these are constants in our character. But we have always understood that when times change, so must we; that fidelity to our founding principles requires new responses to new challenges; that preserving our individual freedoms ultimately requires

collective action. For the American people can no more meet the demands of today's world by acting alone than American soldiers could have met the forces of fascism or communism with muskets and militias. No single person can train all the math and science teachers we'll need to equip our children for the future, or build the roads and networks and research labs that will bring new jobs and businesses to our shores. Now more than ever, we must do these things together, as one nation and one people. This generation of Americans has been tested by crises that steeled our resolve and proved our resilience. A decade of war is now ending. An economic recovery has begun. America's possibilities are limitless, for we possess all the qualities that this world without boundaries demands: youth and drive; diversity and openness; an endless capacity for risk and a gift for reinvention. My fellow Americans, we are made for this moment and we will seize it - so long as we seize it together. For we, the people, understand that our country cannot succeed when a shrinking few do very well and a growing many barely make it. We believe that America's prosperity must rest upon the broad shoulders of a rising middle class. We know that America thrives when every person can find independence and pride in their work; when the wages of honest labor liberate families from the brink of hardship. We are true to our creed when a little girl born into the bleakest poverty knows that she has the same chance to succeed as anybody else, because she is an American; she is free and she is equal, not just in the eyes of God, but also in our own. We understand that outworn programs are inadequate to the needs of our time. So we must harness new ideas and technology to remake our government, revamp our Tax Code, reform our schools, and empower our citizens with the skills they need to work harder, learn more, reach higher. But while the means will change, our purpose endures: a nation that rewards the effort and determination of every single American. That is what this moment requires. That is what will give real meaning to our creed. We, the people, still believe that every citizen deserves a basic measure of security and dignity. We must make the hard choices to reduce the cost of health care and the size of our deficit. But we reject the belief that America must choose between caring for the generation that built this country and investing in the generation that will build its future. For we remember the lessons of our past, when twilight years were spent in poverty and parents of a child with a disability had nowhere to turn. We do not believe that in this country freedom is reserved for the lucky, or happiness for the few. We recognize that no matter how responsibly we live our lives, any one of us at any time may face a job loss or a sudden illness or a home swept away in a terrible storm. The commitments we make to each other through Medicare and Medicaid and Social Security, these things do not sap our initiative, they strengthen us. They do not make us a nation of takers; they free us to take the risks that make this country great. We, the people, still believe that our obligations as Americans are not just to ourselves, but to all posterity. We will respond to the threat of climate change, knowing that the failure to do so would betray our children and future generations. Some may still deny the overwhelming judgment of science, but none can avoid the devastating impact of raging fires and crippling drought and more powerful storms. The path towards sustainable energy sources will be long and sometimes difficult. But America cannot resist this transition, we must lead it. We cannot cede to other nations the technology that will power new jobs and new industries, we must claim its promise. That's how we will maintain our economic vitality and our national treasure - our forests and waterways, our crop lands and snow-capped peaks. That is how we will preserve our planet, commanded to our care by God. That's what will lend meaning to the creed our fathers once declared. We, the

people, still believe that enduring security and lasting peace do not require perpetual war. Our brave men and women in uniform, tempered by the flames of battle, are unmatched in skill and courage. Our citizens, seared by the memory of those we have lost, know too well the price that is paid for liberty. The knowledge of their sacrifice will keep us forever vigilant against those who would do us harm. But we are also heirs to those who won the peace and not just the war; who turned sworn enemies into the surest of friends - and we must carry those lessons into this time as well. We will defend our people and uphold our values through strength of arms and rule of law. We will show the courage to try and resolve our differences with other nations peacefully - not because we are naive about the dangers we face, but because engagement can more durably lift suspicion and fear. America will remain the anchor of strong alliances in every corner of the globe. And we will renew those institutions that extend our capacity to manage crisis abroad, for no one has a greater stake in a peaceful world than its most powerful nation. We will support democracy from Asia to Africa, from the Americas to the Middle East, because our interests and our conscience compel us to act on behalf of those who long for freedom. And we must be a source of hope to the poor, the sick, the marginalized, the victims of prejudice - not out of mere charity, but because peace in our time requires the constant advance of those principles that our common creed describes: tolerance and opportunity, human dignity and justice. We, the people, declare today that the most evident of truths - that all of us are created equal - is the star that guides us still; just as it guided our forebears through Seneca Falls and Selma and Stonewall; just as it guided all those men and women, sung and unsung, who left footprints along this great Mall, to hear a preacher say that we cannot walk alone; to hear a King proclaim that our individual freedom is inextricably bound to the freedom of every soul on Earth. It is now our generation's task to carry on what those pioneers began. For our journey is not complete until our wives, our mothers and daughters can earn a living equal to their efforts. Our journey is not complete until our gay brothers and sisters are treated like anyone else under the law - for if we are truly created equal, then surely the love we commit to one another must be equal as well. Our journey is not complete until no citizen is forced to wait for hours to exercise the right to vote. Our journey is not complete until we find a better way to welcome the striving, hopeful immigrants who still see America as a land of opportunity - until bright young students and engineers are enlisted in our workforce rather than expelled from our country. Our journey is not complete until all our children, from the streets of Detroit to the hills of Appalachia, to the quiet lanes of Newtown, know that they are cared for and cherished and always safe from harm. That is our generation's task - to make these words, these rights, these values of life and liberty and the pursuit of happiness real for every American. Being true to our founding documents does not require us to agree on every contour of life. It does not mean we all define liberty in exactly the same way or follow the same precise path to happiness. Progress does not compel us to settle centuries-long debates about the role of government for all time, but it does require us to act in our time. For now decisions are upon us and we cannot afford delay. We cannot mistake absolutism for principle or substitute spectacle for politics or treat name-calling as reasoned debate. We must act, knowing that our work will be imperfect. We must act, we must act knowing that today's victories will be only partial and that it will be up to those who stand here in 4 years and 40 years and 400 years hence to advance the timeless spirit once conferred to us in a spare Philadelphia hall. My fellow Americans, the oath I have sworn before you today, like the one recited by others who serve in this Capitol, was an

oath to God and country, not party or faction. And we must faithfully execute that pledge during the duration of our service. But the words I spoke today are not so different from the oath that is taken each time a soldier signs up for duty or an immigrant realizes her dream. My oath is not so different from the pledge we all make to the flag that waves above and that fills our hearts with pride. They are the words of citizens and they represent our greatest hope. You and I, as citizens, have the power to set this country's course. You and I, as citizens, have the obligation to shape the debates of our time - not only with the votes we cast, but with the voices we lift in defense of our most ancient values and enduring ideals. Let us, each of us, now embrace with solemn duty and awesome joy what is our lasting birthright. With common effort and common purpose, with passion and dedication, let us answer the call of history and carry into an uncertain future that precious light of freedom. Thank you. God bless you, and may He forever bless these United States of America.

Paragraph atomization

The unit of analysis is the paragraph: a comment spans several topics, and a topic model’s “document” is whatever unit the DTM rows are. Merging is **direction-aware**, because short blocks come in two kinds that belong in opposite directions. Headings (at most 12 tokens, no sentence-ending punctuation) merge **forward** into the paragraph they introduce, since a naive rule would glue a subtitle’s words onto the *preceding* paragraph and misfile its topic signal; a trailing heading falls back to the last body paragraph. Short body fragments that do end like sentences merge **backward** into what they continue, absorbing codas, salutations, and signature lines.

```
count_tokens <- function(x) str_count(x, "\\S+")

paragraphs <- cache_csv(p_out("paragraphs.csv"), {
  # PASS 1 (forward): rev(cumsum(rev(!heading))) gives each body block an id
  # that decreases down the document and gives a heading the id of the NEXT
  # body block; trailing headings get 0, folded in by pmax(1L).
  corpus_raw |>
    dplyr::mutate(para_raw = str_split(body_english, tx$para_split)) |>
    dplyr::select(atom_id, Year, para_raw) |>
    tidyr::unnest(para_raw) |>
    dplyr::mutate(para_raw = str_squish(para_raw)) |>
    dplyr::filter(para_raw != "") |>
    dplyr::group_by(atom_id, Year) |>
    dplyr::mutate(
      heading = count_tokens(para_raw) <= tx$heading_max_tokens &
        !str_detect(para_raw, "[.!?][\"\\\"\\\\\\\\]*$"),
      fwd      = rev(cumsum(rev(!heading))) |> pmax(1L)) |>
    dplyr::group_by(atom_id, Year, fwd) |>
    dplyr::summarise(para_raw = str_c(para_raw, collapse = " "),
      .groups = "drop") |>
    dplyr::arrange(atom_id, Year, dplyr::desc(fwd)) |>
  # PASS 2 (backward): short body fragments join the preceding paragraph.
  dplyr::group_by(atom_id, Year) |>
```

```

dplyr::mutate(full = count_tokens(para_raw) >= tx$para_min_tokens,
              blk = cummax(cumsum(full) * full) |> pmax(1L)) |>
dplyr::group_by(atom_id, Year, blk) |>
dplyr::summarise(text = str_c(para_raw, collapse = " "),
                  .groups = "drop_last") |>
dplyr::mutate(para_id = dplyr::row_number(),
              para_uid = str_c(atom_id, "-", sprintf("%02d", para_id)),
              n_tokens = count_tokens(text)) |>
dplyr::ungroup() |>
dplyr::filter(n_tokens >= tx$para_min_tokens) |>
dplyr::select(para_uid, atom_id, para_id, Year, n_tokens, text)
})
glue::glue("{nrow(paragraphs)} paragraphs from ",
            "{dplyr::n_distinct(paragraphs$atom_id)} comments")

```

1573 paragraphs from 60 comments

Corpus description

```

paragraphs |>
  dplyr::count(Year) |>
  ggplot(aes(Year, n)) +
  geom_col(fill = viridis(1, begin = 0.35)) +
  labs(x = NULL, y = "paragraphs")

```

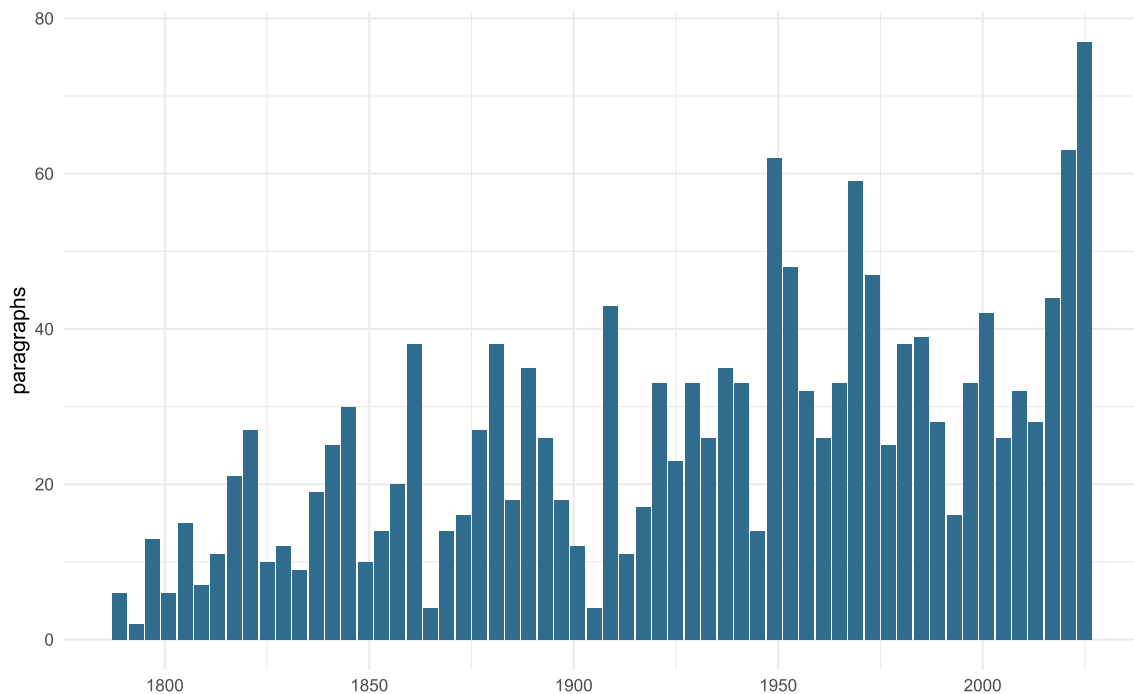


Figure 1: Paragraph volume by year: the denominator for every time-indexed estimate. Thin years mean the prevalence spline is interpolating and single-year claims are weak.

```
tokens <- paragraphs |>
  tidytext::unnest_tokens(word, text) |>
  dplyr::anti_join(tidytext::get_stopwords(), by = "word") |>
  dplyr::filter(str_detect(word, "[a-z]"))
```

Document-term matrix

Rare terms (under 5 paragraphs) are trimmed to stabilize estimation. `meta` is built from the surviving `dfm` rownames, so alignment with the model survives any dropped document. At work, check `nfeat(dfm_para)`: a vocabulary in the low thousands is comfortable at this scale, and a much larger one argues for a higher floor.

```
dfm_para <- tokens |>
  dplyr::count(para_uid, word) |>
  tidytext::cast_dfm(para_uid, word, n) |>
  quantda::dfm_trim(min_docfreq = tx$min_docfreq)
stm_in <- quantda::convert(dfm_para, to = "stm")
# as.data.frame, not a tibble: stm subsets metadata with base [ , ] semantics
# when building the prevalence model matrix, and a tibble returns a column
# rather than a vector there ("Error creating model matrix").
meta <- tibble(para_uid = rownames(dfm_para)) |>
  dplyr::left_join(paragraphs, by = "para_uid") |>
  as.data.frame()
stopifnot("every dfm row must match a paragraph" = !anyNA(meta$Year),
  "meta must align with the stm documents" =
    nrow(meta) == length(stm_in$documents))

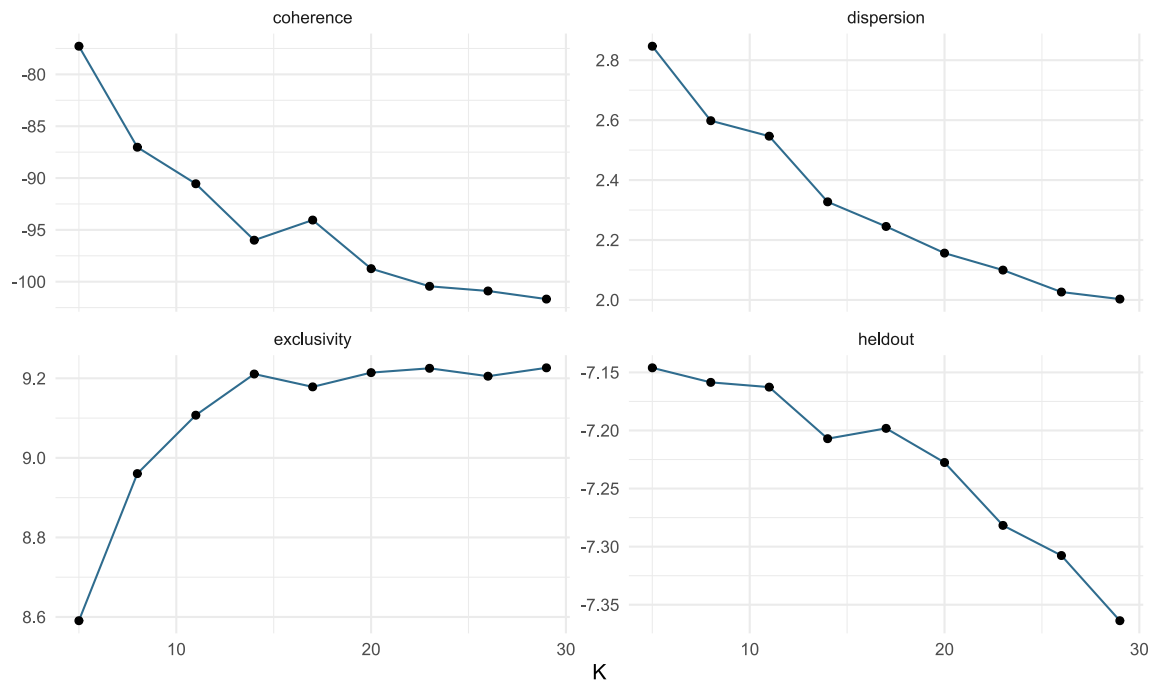
# Spline basis as ORDINARY COLUMNS, so no function call ever appears inside a
# formula. stm builds its prevalence model matrix by evaluating the formula
# against the metadata, and a call there (s(), ns(), or anything namespaced)
# is the documented cause of "Error creating model matrix". prev_form then
# reads ~ yr1 + yr2 + ..., which is also what estimateEffect uses below.
bs <- splines::ns(meta$Year, df = tx$spline_df)
colnames(bs) <- str_c("yr", seq_len(ncol(bs)))
meta <- cbind(meta, as.data.frame(bs))
prev_rhs <- str_c(colnames(bs), collapse = " + ") # "yr1 + yr2 + ..."
prev_form <- as.formula(str_c("~ ", prev_rhs))
```

Choosing K

`stm`'s own parallelism forks, which is unavailable on Windows, so the held-out grid is parallelized explicitly with `furrr`. Read the panels together: take the smallest `K` where held-out likelihood has flattened and residual dispersion has stopped falling, then break ties on the coherence-exclusivity frontier. If the frontier is still improving at the top of the grid, widen `tx$K_grid` rather than

extrapolating. Semantic coherence is $\sum_{i < j} \log\{(D(v_i, v_j) + 1)/D(v_i)\}$ over each topic's top words, with the denominator the higher-ranked word.

```
if (isTRUE(tx$run_searchK)) {
  future::plan(future::multisession, workers = tx$workers)
  ho <- stm::make.heldout(stm_in$documents, stm_in$vocab, seed = tx$seed)
  metrics <- cache_csv(p_out("searchK_metrics.csv"), {
    furrr::future_map(tx$K_grid, function(k) {
      fit <- stm::stm(ho$documents, ho$vocab, K = k, init.type = "Spectral",
        prevalence = prev_form, data = meta, verbose = FALSE)
      tibble(K = k,
        heldout = stm::eval.heldout(fit, ho$missing)$expected.heldout,
        dispersion = stm::checkResiduals(fit, ho$documents)$dispersion,
        coherence = mean(stm::semanticCoherence(fit, ho$documents)),
        exclusivity = mean(stm::exclusivity(fit)))
    }, .options = furrr::furrr_options(seed = tx$seed)) |> purrr::list_rbind()
  })
  future::plan(future::sequential)
  metrics |>
    tidyr::pivot_longer(-K) |>
    ggplot(aes(K, value)) + geom_line(color = viridis(1, begin = 0.35)) +
    geom_point() + facet_wrap(~ name, scales = "free_y") + labs(y = NULL)
} else {
  if (is.null(tx$K))
    stop("tx$K is NULL and tx$run_searchK is FALSE. Re-run the config chunk, ",
      "or set tx$K before continuing.")
  glue::glue("searchK skipped (tx$run_searchK = FALSE); K fixed at {tx$K}")
}
```



Structural topic model

Year enters the prevalence model as a spline (df 5), so topic attention is allowed to drift over the period instead of being averaged away; spectral initialization is deterministic given the data, removing initialization luck from a fit everything downstream depends on.

```
stm_fit <- cache_rds(p_out("stm_fit.rds"),
  stm::stm(stm_in$documents, stm_in$vocab, K = tx$K, init.type = "Spectral",
    prevalence = prev_form, data = meta, verbose = FALSE))

theta <- stm_fit$theta
colnames(theta) <- str_c("topic_", seq_len(tx$K))
paragraph_theta <- meta |> dplyr::bind_cols(tibble::as_tibble(theta))
```

Modal topic assignment and its diagnostics

Each paragraph is assigned its **modal** topic, so topics partition the corpus (mutually exclusive, collectively exhaustive) and can serve as sampling strata in Script 2. This is the same move as modal class assignment in latent class analysis, and it inherits the same caveat: a modal assignment is an estimate, not an observation, and treating it as observed pushes classification error downstream. Two LCA diagnostics report how much error to expect. **Entropy** R^2 summarizes overall assignment sharpness on a 0-1 scale; **AvePP**, the average posterior probability among a topic's own assigned paragraphs, does it per topic, with about 0.7 the conventional floor. Unlike LCA, STM supplies no classification error matrix over modal assignment, so no BCH-style cor-

rection is available here; the honest substitutes are the min_theta scope floor in Script 2 and its sensitivity analysis.

```
paragraph_theta <- paragraph_theta |>
  dplyr::mutate(topic      = max.col(theta, ties.method = "first"),
               theta_max = theta[cbind(dplyr::row_number(), topic)])

ent <- -sum(theta * log(theta + 1e-12))
entropy_r2 <- 1 - ent / (nrow(theta) * log(tx$K))

topic_summary <- tibble(
  topic = seq_len(tx$K),
  prevalence = colMeans(theta),
  frex = apply(stm::labelTopics(stm_fit, n = 8)$frex, 1, str_c,
               collapse = ", "),
  prob = apply(stm::labelTopics(stm_fit, n = 8)$prob, 1, str_c,
               collapse = ", ")) |>
  dplyr::left_join(
    paragraph_theta |>
      dplyr::group_by(topic) |>
      dplyr::summarise(n paras = dplyr::n(), avepp = mean(theta_max)),
    by = "topic")

readr::write_csv(topic_summary, p_out("topic_summary.csv"))
readr::write_csv(paragraph_theta, p_out("paragraph_theta.csv"))

glue::glue("Entropy R2 = {round(entropy_r2, 3)}; topics with AvePP < 0.7: ",
           "{sum(topic_summary$avepp < 0.7)} of {tx$K}")
```

Entropy R2 = 0.451; topics with AvePP < 0.7: 12 of 12

```
topic_summary |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  dplyr::select(topic, prevalence, n paras, avepp) |>
  tbl_paged(caption = "Topics by prevalence, with assigned paragraph counts and
AvePP (below 0.7 means fuzzy modal assignment).")
```

Topics by prevalence, with assigned paragraph counts and AvePP (below 0.7 means fuzzy modal assignment).

topic	prevalence	n paras	avepp
8	0.145	289	0.504
1	0.106	181	0.525
6	0.096	140	0.546
11	0.095	125	0.484

topic	prevalence	n_paras	avepp
4	0.093	128	0.524
3	0.080	140	0.609
7	0.077	108	0.633
10	0.070	127	0.540
9	0.069	107	0.593
5	0.062	74	0.512
12	0.055	57	0.509
2	0.053	97	0.523

```
topic_summary |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  dplyr::mutate(ttl = str_c("Topic ", topic)) |>
  tbl_records("ttl", c("frex", "prob"),
    caption = "Topic terms: the evidence for checking each label.")
```

Topic terms: the evidence for checking each label.

Topic 8 - frex: americans, together, america's, story, challenges, voices, promise, moment - prob: us, america, let, americans, time, together, today, government

Topic 1 - frex: peoples, peace, belief, world, enemies, help, seek, nations - prob: world, peace, nations, free, freedom, can, america, know

Topic 6 - frex: faith, truth, destiny, fathers, prayer, light, self, liberty - prob: us, faith, shall, let, liberty, people, god, freedom

Topic 11 - frex: new, century, ago, years, four, last, millions, first - prob: new, world, years, now, first, nation, century, history

Topic 4 - frex: thank, children, believe, think, bless, say, families, simple - prob: can, must, country, believe, now, children, make, work

Topic 3 - frex: revenue, tariff, business, expenditures, taxation, officers, amendment, treasury - prob: public, government, laws, shall, congress, revenue, business, may

Topic 7 - frex: president, oath, trust, office, chief, presence, duties, vice - prob: citizens, fellow, president, duties, office, people, shall, trust

Topic 10 - frex: constitution, general, powers, confederacy, union, authority, territories, grant - prob: government, constitution, states, union, upon, people, rights, powers

Topic 9 - frex: foreign, treaty, trade, countries, relations, navy, commerce, naval - prob: foreign, war, states, nations, united, peace, great, policy

Topic 5 - frex: order, agencies, clear, essential, determined, cooperation, full, courts - prob: must, government, can, order, people, national, full, justice

Topic 12 - frex: industrial, race, citizenship, things, save, problems, feeling, capacity - prob: upon, great, good, people, government, things, nation, without

Topic 2 - frex: suffrage, virtue, measures, sentiments, south, devoted, opinion, southern - prob: people, may, country, government, every, can, spirit, power

Topic codebook

The codebook is written **once**, after the final STM, and is the naming record from here on. If `topic_codebook.csv` already exists on disk, this section reads it and makes no API calls: the file is assumed to carry your edits, and re-drafting would overwrite them. Delete the file to redraft from scratch.

Two passes, deliberately shaped:

Pass 1, one call per topic with the smaller model (label step). Each call sees only that topic's own evidence: its FREX terms, its highest-probability terms, and its three highest-theta paragraphs. It is blind to every other topic, to Year, and to stance. That blindness is the point: a model shown all topics at once starts fitting a narrative across them, and labels stop being traceable to the evidence a human can check them against.

Pass 2, one manager call with the larger model (manager step). Blind drafting has one predictable failure: two topics can come back with near-duplicate labels even though their term distributions differ. The manager sees all K labels with their FREX terms and resolves collisions. Its mandate is deliberately narrow, stated in the prompt and enforced by the schema: it may **rename only**. It cannot merge topics, split them, change a description, or reorder anything, because the topic structure is the statistical model's output and an LLM should not be permitted to override it.

The proposition column is written empty here. Script 2 fills it for the one topic it processes, since propositions are stance targets and you only need them for the topics you actually analyze.

```
cb_path <- p_out("topic_codebook.csv")

if (file.exists(cb_path)) {
  codebook <- readr::read_csv(cb_path, show_col_types = FALSE)
  writeLines(glue::glue("Existing codebook found ({nrow(codebook)} topics); \\
                        no LLM calls made. Delete {basename(cb_path)} to \\
                        redraft."))
} else {
  ev <- topic_summary |>
  dplyr::left_join(
    paragraph_theta |>
    dplyr::group_by(topic) |>
    dplyr::slice_max(theta_max, n = 3) |>
    dplyr::summarise(ex = str_c(str_trunc(text, 300), collapse = "\\n- "),
```

```

      .groups = "drop"),
    by = "topic")

# ---- Pass 1: one blind call per topic -----
label_sys <- build_prompt(
  role = str_c("You are a survey methodologist who names statistical topics ",
    "so colleagues can use them."),
  task = str_c("Read this topic's distinctive terms and example paragraphs. ",
    "Write a short label and one factual sentence describing what ",
    "these paragraphs are about. Before answering, check that ",
    "every word of your description is supported by the evidence ",
    "shown."),
  rules = c("Use only the terms and paragraphs shown.",
    "Do not infer stance, sentiment, intent, or who is speaking.",
    "Do not invent statistics, dates, places, or named entities.",
    "If the evidence is mixed, describe the mixture plainly."),
  output = "A label of at most 4 words and a one-sentence description.")
label_schema <- ellmer::type_object(
  label = ellmer::type_string("Topic name, at most 4 words."),
  description = ellmer::type_string("One factual sentence. "))
label_chat <- make_chat("label", label_sys, tx$steps, tx$home)

drafts <- purrr::map(seq_len(nrow(ev)), function(i) {
  r <- call_one(label_chat,
    str_c("Distinctive (FREX) terms: ", ev$frex[i],
      "\nHighest-probability terms: ", ev$prob[i],
      "\nExample paragraphs:\n- ", ev$ex[i]), label_schema)
  tibble(topic = ev$topic[i],
    label = r$label %||% NA_character_,
    description = r$description %||% NA_character_)
}) |> purrr::list_rbind()
if (anyNA(drafts$label))
  warning(sum(is.na(drafts$label)), " topic label calls failed; those rows ",
    "are blank in the codebook and need a name by hand.")

# ---- Pass 2: manager resolves duplicate labels, renaming ONLY -----
mgr_sys <- build_prompt(
  role = str_c("You are a senior methodologist reviewing a topic codebook ",
    "for clarity before it goes to coders."),
  task = str_c("You are shown every topic's draft label with its ",
    "distinctive terms. Some labels are near-duplicates of each ",
    "other even though the topics differ. Return a label for ",
    "EVERY topic: unchanged where it is already distinct, and ",
    "revised only where two or more labels would be confused. ",
    "Before answering, check that no two returned labels are ",

```

```

        "the same and that every topic number appears exactly once."),
rules = c("Rename only. Never merge, split, drop, or reorder topics.",
        "Return the same number of topics you were given.",
        "Base any rename on the distinctive terms shown, nothing else.",
        "Keep labels to at most 4 words.",
        "Do not write descriptions; they are not yours to change."),
output = "One label per topic number.")
mgr_schema <- ellmer::type_object(
  topics = ellmer::type_array(
    items = ellmer::type_object(
      topic = ellmer::type_integer("The topic number, unchanged."),
      label = ellmer::type_string("Final label, at most 4 words.")))
mgr_prompt <- str_c(
  "Draft labels with their distinctive terms:\n",
  str_c("- topic ", drafts$topic, ": \"", drafts$label, "\" | terms: ",
    ev$frex[match(drafts$topic, ev$topic)], collapse = "\n"))
mgr <- call_one(make_chat("manager", mgr_sys, tx$steps, tx$home),
  mgr_prompt, mgr_schema)

# The manager is advisory: its output is applied ONLY if it returns exactly
# one label per topic with no topics invented or lost. Any deviation and the
# blind drafts stand, because a malformed rename must never reshape the
# codebook silently.
final <- if (!is.null(mgr) && length(mgr$topics) == nrow(drafts)) {
  renamed <- purrr::map(mgr$topics, ~ tibble(topic = as.integer(.x$topic),
    label_new = .x$label)) |>
    purrr::list_rbind()
  if (setequal(renamed$topic, drafts$topic)) {
    drafts |> dplyr::left_join(renamed, by = "topic") |>
      dplyr::mutate(label = dplyr::coalesce(label_new, label)) |>
      dplyr::select(-label_new)
  } else drafts
} else drafts
n_renamed <- sum(final$label != drafts$label, na.rm = TRUE)

codebook <- final |>
  dplyr::mutate(proposition = NA_character_) |> # Script 2 fills this
  dplyr::left_join(topic_summary |> dplyr::select(topic, prevalence, avepp),
    by = "topic") |>
  dplyr::arrange(topic)
readr::write_csv(codebook, cb_path)
writeLines(glue::glue("Codebook drafted: {nrow(codebook)} topics, \\
  {n_renamed} renamed by the manager. EDIT \\
  {basename(cb_path)} before running Script 2; it will \\
  not be redrafted while the file exists."))
}

```

Existing codebook found (12 topics); no LLM calls made. Delete topic_codebook.csv to redraft.

```
codebook |>
  dplyr::mutate(ttl = str_c("Topic ", topic, ": ", label)) |>
  tbl_records("ttl", "description",
    caption = "Topic codebook. Labels drafted blind per topic, then
deduplicated by a rename-only manager pass. Edit the CSV; nothing downstream
redrafts it.")
```

Topic codebook. Labels drafted blind per topic, then deduplicated by a rename-only manager pass. Edit the CSV; nothing downstream redrafts it.

Topic 1: World Peace - description: These paragraphs are about promoting peace and freedom in the world among nations.

Topic 2: Government Opinion - description: These paragraphs discuss the role of government and public opinion in relation to the country's institutions and virtue.

Topic 3: Government Revenue - description: These paragraphs are about the relationship between government revenue, business, and taxation, including the impact of tariff revisions and expenditures on the economy.

Topic 4: National Unity - description: These paragraphs are about the country and its people, including families and children, and what they can and must do to succeed and be treated equally under the law.

Topic 5: Government Order - description: These paragraphs are about the importance of effective government and political parties at home in order to achieve continued success abroad and domestically.

Topic 6: National Faith - description: These paragraphs are about a nation seeking blessings and guidance from God while striving for unity, strength, and liberty.

Topic 7: Presidential Address - description: These paragraphs are about a president assuming office and discussing their duties and trust with the citizens.

Topic 8: US Unity - description: These paragraphs are about Americans acting together to meet challenges and fulfill America's promise.

Topic 9: Foreign Policy - description: These paragraphs are about the conduct of foreign policy and relations with other nations.

Topic 10: Government Powers - description: These paragraphs are about the distribution of powers between the United States government and the states, as defined by the Constitution.

Topic 11: New Century - description: The paragraphs describe changes in the world over the last few centuries, including advancements in communication and commerce, and the impact of these changes on the nation and the world.

Topic 12: National Problems - description: These paragraphs discuss the great aspects of life, wealth, and industry, as well as the problems and waste that have come with them.

Representative paragraphs

```
top4 <- topic_summary |> dplyr::slice_max(prevalence, n = 4) |>
  dplyr::pull(topic)
th <- stm::findThoughts(stm_fit, texts = meta$para_uid, n = 3, topics = top4)
tibble(topic = rep(top4, each = 3), para_uid = unlist(th$docs)) |>
  dplyr::left_join(paragraphs |> dplyr::select(para_uid, text),
    by = "para_uid") |>
  dplyr::mutate(ttl = str_c("Topic ", topic, " | ", para_uid)) |>
  tbl_records("ttl", "text",
    caption = "Three highest-theta paragraphs for the four most
prevalent topics.")
```

Three highest-theta paragraphs for the four most prevalent topics.

Topic 8 | 1989-Bush-20 - text: To my friends - and yes, I do mean friends - in the loyal opposition - and yes, I mean loyal: I put out my hand. I am putting out my hand to you, Mr. Speaker. I am putting out my hand to you Mr. Majority Leader. For this is the thing: This is the age of the offered hand. We can't turn back clocks, and I don't want to. But when our fathers were young, Mr. Speaker, our differences ended at the water's edge. And we don't wish to turn back time, but when our mothers were young, Mr. Majority Leader, the Congress and the Executive were capable of working together to produce a budget on which this nation could live. Let us negotiate soon and hard. But in the end, let us produce. The American people await action. They didn't send us here to bicker. They ask us to rise above the merely partisan. "In crucial things, unity" - and this, my friends, is crucial.

Topic 8 | 2021-Biden-24 - text: This is our historic moment of crisis and challenge, and unity is the path forward. And, we must meet this moment as the United States of America. If we do that, I guarantee you, we will not fail. We have never, ever, ever failed in America when we have acted together. And so today, at this time and in this place, let us start afresh. All of us. Let us listen to one another. Hear one another. See one another. Show respect to one another. Politics need not be a raging fire destroying everything in its path. Every disagreement doesn't have to be a cause for total war.

Topic 8 | 2009-Obama-20 - text: Recall that earlier generations faced down fascism and communism not just with missiles and tanks, but with the sturdy alliances and enduring convictions. They understood that our power alone cannot protect us, nor does it entitle us to do as we please. Instead, they knew that our power grows through its prudent use; our security emanates from the justness of our cause, the force of our example, the tempering qualities of humility and restraint.

Topic 1 | 1953-Eisenhower-35 - text: (6) Recognizing economic health as an indispensable basis of military strength and the free world's peace, we shall strive to foster everywhere, and to practice ourselves, policies that encourage productivity and profitable trade. For the impoverishment of any single people in the world means danger to the well-being of all other peoples.

Topic 1 | 1921-Harding-09 - text: America is ready to encourage, eager to initiate, anxious to participate in any seemly program likely to lessen the probability of war, and promote that broth-

erhood of mankind which must be God's highest conception of human relationship. Because we cherish ideals of justice and peace, because we appraise international comity and helpful relationship no less highly than any people of the world, we aspire to a high place in the moral leadership of civilization, and we hold a maintained America, the proven Republic, the unshaken temple of representative democracy, to be not only an inspiration and example, but the highest agency of strengthening good will and promoting accord on both continents.

Topic 1 | 1949-Truman-15 - text: Communism holds that the world is so deeply divided into opposing classes that war is inevitable. Democracy holds that free nations can settle differences justly and maintain lasting peace.

Topic 6 | 1853-Pierce-04 - text: In this our country has, in my judgment, thus far fulfilled its highest duty to suffering humanity. It has spoken and will continue to speak, not only by its words, but by its acts, the language of sympathy, encouragement, and hope to those who earnestly listen to tones which pronounce for the largest rational liberty. But after all, the most animating encouragement and potent appeal for freedom will be its own history - its trials and its triumphs. Preeminently, the power of our advocacy reposes in our example; but no example, be it remembered, can be powerful for lasting good, whatever apparent advantages may be gained, which is not based upon eternal principles of right and justice. Our fathers decided for themselves, both upon the hour to declare and the hour to strike. They were their own judges of the circumstances under which it became them to pledge to each other "their lives, their fortunes, and their sacred honor" for the acquisition of the priceless inheritance transmitted to us. The energy with which that great conflict was opened and, under the guidance of a manifest and beneficent Providence the uncomplaining endurance with which it was prosecuted to its consummation were only surpassed by the wisdom and patriotic spirit of concession which characterized all the counsels of the early fathers.

Topic 6 | 1957-Eisenhower-03 - text: Before all else, we seek, upon our common labor as a nation, the blessings of Almighty God. And the hopes in our hearts fashion the deepest prayers of our whole people. May we pursue the right - without self-righteousness. May we know unity - without conformity. May we grow in strength - without pride in self.

Topic 6 | 1905-Roosevelt-04 - text: Yet, after all, though the problems are new, though the tasks set before us differ from the tasks set before our fathers who founded and preserved this Republic, the spirit in which these tasks must be undertaken and these problems faced, if our duty is to be well done, remains essentially unchanged. We know that self-government is difficult. We know that no people needs such high traits of character as that people which seeks to govern its affairs aright through the freely expressed will of the freemen who compose it. But we have faith that we shall not prove false to the memories of the men of the mighty past. They did their work, they left us the splendid heritage we now enjoy. We in our turn have an assured confidence that we shall be able to leave this heritage unwasted and enlarged to our children and our children's children. To do so we must show, not merely in great crises, but in the everyday affairs of life, the qualities of practical intelligence, of courage, of hardihood, and endurance, and above all the power of devotion to a lofty ideal, which made great the men who founded this Republic in the days

of Washington, which made great the men who preserved this Republic in the days of Abraham Lincoln.

Topic 11 | 1993-Clinton-04 - text: When George Washington first took the oath I have just sworn to uphold, news travelled slowly across the land by horseback, and across the ocean by boat. Now the sights and sounds of this ceremony are broadcast instantaneously to billions around the world. Communications and commerce are global. Investment is mobile. Technology is almost magical, and ambition for a better life is now universal.

Topic 11 | 1993-Clinton-03 - text: Today, a generation raised in the shadows of the Cold War assumes new responsibilities in a world warmed by the sunshine of freedom, but threatened still by ancient hatreds and new plagues. Raised in unrivalled prosperity, we inherit an economy that is still the world's strongest, but is weakened by business failures, stagnant wages, increasing inequality, and deep divisions among our own people.

Topic 11 | 1965-Johnson-19 - text: Change and the Union The third article was union. To those who were small and few against the wilderness, the success of liberty demanded the strength of union. Two centuries of change have made this true again.

Prevalence over time

```
stm_year_effects <- cache_rds(p_out("stm_year_effects.rds"),
  stm::estimateEffect(
    as.formula(str_c("1:", tx$K, " ~ ", prev_rhs)),
    stm_fit, metadata = meta, uncertainty = "Global"))
```

Expected topic prevalence is predicted across the observed year range from the fitted effect object, with 95 percent intervals. Two details make the prediction correct rather than merely plausible. The spline basis is rebuilt at the prediction grid with `predict(bs, ...)`, which reuses the **original knots**; calling `ns()` afresh on the grid would place knots differently and mis-predict silently. And the intervals come from the simulation draws `estimateEffect` already stores under `uncertainty = "Global"`, so they carry the topic-model uncertainty rather than regression standard errors alone.

```
# Prediction grid: every observed year, with the basis evaluated at the
# ORIGINAL knots so the fitted coefficients apply.
grid <- tibble(Year = sort(unique(meta$Year)))
Xg <- cbind(1, predict(bs, newx = grid$Year))

# estimateEffect stores per-topic simulation draws of the coefficient vector
# in $parameters[[k]][[j]]$est. Predicting each draw and taking quantiles
# propagates both the regression and the topic-model uncertainty.
draws_for_topic <- function(k) {
  purrr::map(stm_year_effects$parameters[[k]], function(sim)
    as.numeric(Xg %*% sim$est)) |>
  purrr::reduce(rbind) |> matrix(ncol = nrow(grid))
```

```

}
topic_trends <- purrr::map(seq_along(stm_year_effects$topics), function(i) {
  d <- draws_for_topic(i)
  grid |> dplyr::mutate(
    topic = stm_year_effects$topics[i],
    fit   = apply(d, 2, mean),
    lo    = apply(d, 2, quantile, 0.025),
    hi    = apply(d, 2, quantile, 0.975))
}) |> purrr::list_rbind()
readr::write_csv(topic_trends, p_out("topic_trends.csv"))
glue::glue("Predicted prevalence for {dplyr::n_distinct(topic_trends$topic)} \
\
          topics across {nrow(grid)} years written to topic_trends.csv")

```

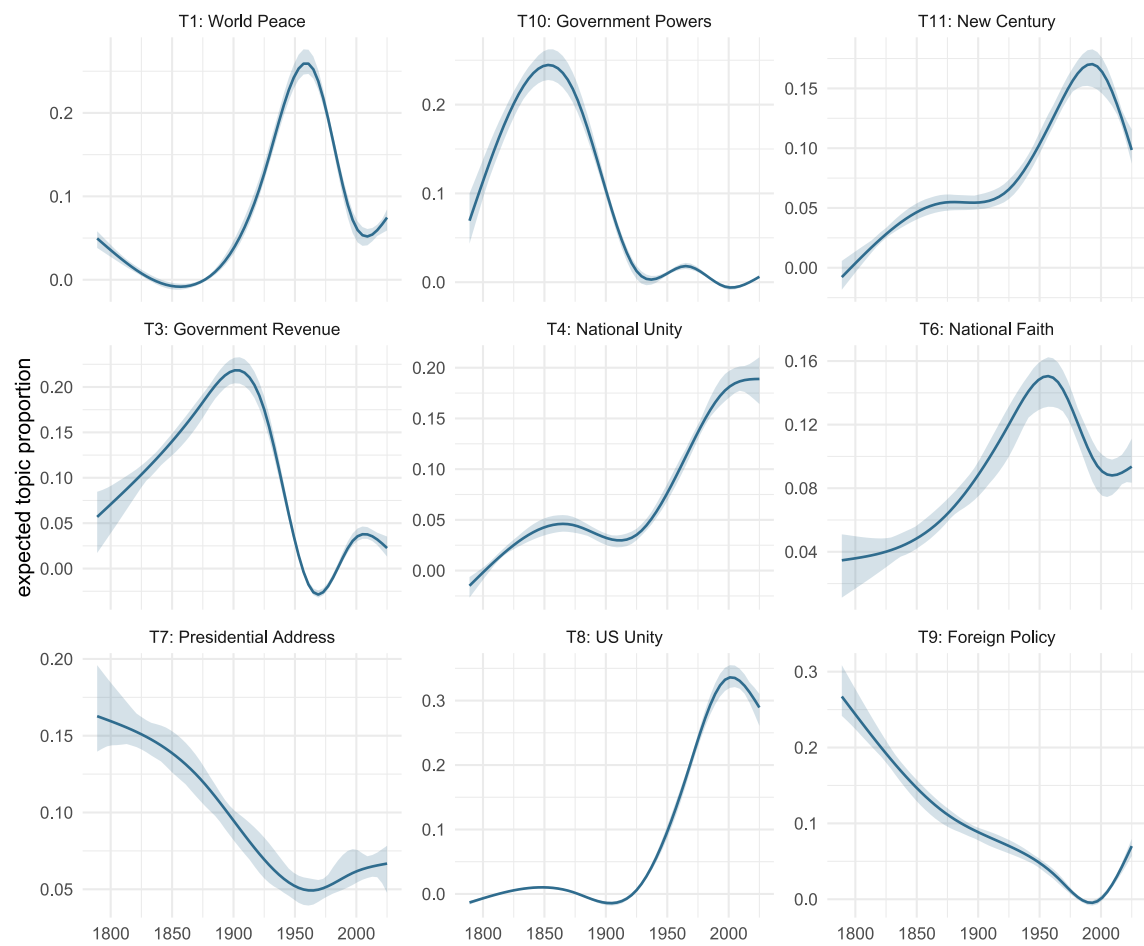
Predicted prevalence for 12 topics across 60 years written to topic_trends.csv

```

# All topics, paged so panels stay legible: 25 facets on one page renders each
# about 1.5 inches wide and the ribbons become unreadable.
trend_panel <- function(d, page, n_pages) {
  ggplot(d, aes(Year, fit)) +
    geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2,
              fill = viridis(1, begin = 0.35)) +
    geom_line(linewidth = 0.7, color = viridis(1, begin = 0.35)) +
    facet_wrap(~ topic_label, scales = "free_y", ncol = 3) +
    labs(x = NULL, y = "expected topic proportion",
         title = glue::glue("Topic prevalence over time ({page} of {n_pages})"),
         subtitle = "Model-based estimates with 95 percent intervals")
}
trend_pages <- topic_trends |>
  dplyr::left_join(topic_summary |> dplyr::select(topic, prevalence),
                  by = "topic") |>
  dplyr::left_join(codebook |> dplyr::select(topic, label), by = "topic") |>
  dplyr::mutate(
    topic_label = str_c("T", topic, ": ", str_trunc(label, 22)),
    page = (dplyr::dense_rank(dplyr::desc(prevalence)) - 1L) %/%
           tx$facets_per_page + 1L) |>
  dplyr::group_split(page)
purrr::iwalk(trend_pages, function(d, i)
  print(trend_panel(d, i, length(trend_pages))))

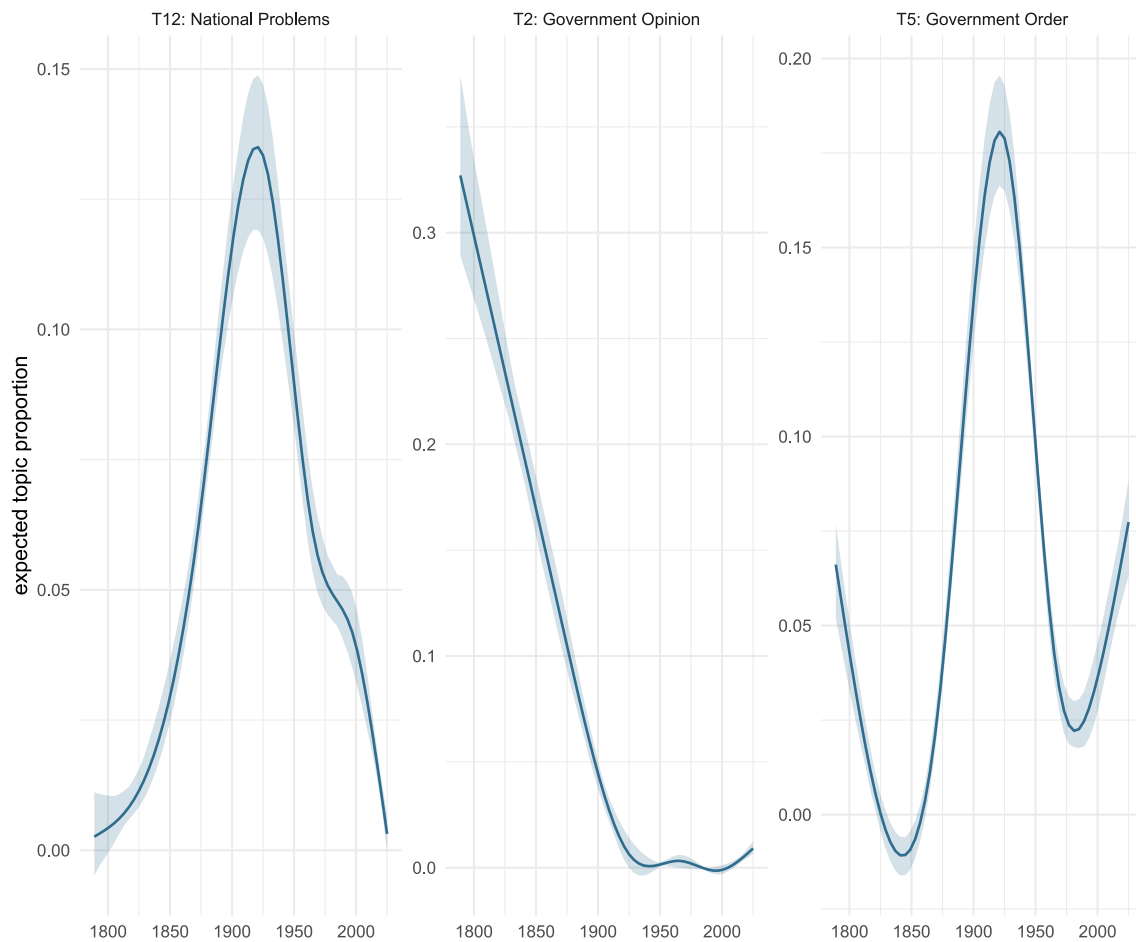
```

Topic prevalence over time (1 of 2)
Model-based estimates with 95 percent intervals



Topic prevalence over time (2 of 2)

Model-based estimates with 95 percent intervals



```
glue::glue("outputs/ written: paragraphs.csv, paragraph_theta.csv (ids, Year, \
\
n_tokens, text, topic_1..K, topic, theta_max), topic_summary.csv, \
topic_codebook.csv (labels; Script 2 adds propositions), topic_trends.csv, \
stm_fit.rds, stm_year_effects.rds. Script 2 reads these.")
```

```
outputs/ written: paragraphs.csv, paragraph_theta.csv (ids, Year, n_tokens,
text, topic_1..K, topic, theta_max), topic_summary.csv, topic_codebook.csv
(labels; Script 2 adds propositions), topic_trends.csv,
stm_fit.rds, stm_year_effects.rds. Script 2 reads these.
```